Vehicle Classification

Transfer learning and Neysan subtype generalization

Transfer learning for Iranian traffic vehicle classification and Neysan subtype generalization

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A B S T R A C T

This project develops an eight-class vehicle image classifier using transfer learning on a CPU. Images from five source folders were pooled, manually reviewed, checked for readability, and deduplicated before creating new splits. The recorded audit contained 3,536 readable files and 3,512 unique files. In phase one, ResNet18, ResNet34, and ResNet50 were compared using a stratified 70% training, 15% validation, and 15% test split. ResNet18 was selected, then evaluated with dropout, balanced batches, and alternative losses. The selected validation model reached 98.86% accuracy and 0.9890 macro-F1. A fixed eight-epoch refit on training plus validation achieved 97.53% accuracy and 0.9746 macro-F1 on 527 held-out test images. In phase two, 243 unique Neysan images were held out while the remaining eight-class pool was split 80/20. A fresh ImageNet-initialized ResNet18 was trained with layer3, layer4, and the classifier unfrozen. After selection and a ten-epoch refit, 228 of 243 Neysan images were predicted as Vanet (93.83%). A confidence threshold of 0.8 flagged 39 images, capturing 10 of 15 errors. All eligible images were scored before review copies were created. The results support the chosen model within these datasets, while highlighting the limits of uncalibrated confidence and a single-subtype test.

Keywords: vehicle classification; ResNet18; transfer learning; Neysan; Vanet; class imbalance; human review.

1. Introduction

The goal is to classify a vehicle image into one of eight practical categories. The project also asks a more specific question: can a model recognize Neysan vehicles as members of the broader Vanet category when Neysan examples are excluded from training? The two phases answer different questions and are reported separately.

The workflow prioritizes readable code, preserved source data, saved checkpoints, and clear evaluation. No detector is trained in this project. The input is an existing vehicle image, and the output is one class with a confidence score.

2. Dataset sources and manual revision ledger

The original sources were dataset1/train, dataset1/test, dataset1/unclean, datasetv2_TrainUclean/train, and datasetv2_TrainUclean/unclean. Their names describe origins only. Neither an old Train folder nor an old Test folder defines the final experimental split. All approved sources were pooled before a new stratified split was created.

Source	Recorded readable files
dataset1/train	399
dataset1/test	399
dataset1/unclean	449
datasetv2_TrainUclean/train	1,533
datasetv2_TrainUclean/unclean	756
Total	3,536

Table 1. Source counts calculated from the saved audit outputs. The second dataset has only five classes in its Unclean source.

The author reported the following manual corrections. Removed images were excluded from the dataset; moved images were relabeled by changing their class folder. The reason for each individual change and the affected filenames were not included in the submitted materials, so this report does not invent those details.

Action	Number	Change in class membership
Remove	2	Taxi
Remove	2	Ambulance
Remove	1	Minibus
Move	109	kamyun to kamyunet
Move	2	ambulance to savari

Table 2. Author-reported manual dataset revision ledger. Five files were removed and 111 files were moved; moving does not reduce the total number of files.

The saved notebooks are the numerical record used for the experiments. The manual ledger is documented alongside that record, rather than subtracted again from the audited totals. No before-and-after file manifest was supplied to reconstruct the exact timing of these edits. Applying the changes a second time would create unsupported counts.

In phase one, NEYSAN, Nysan, and Nissan naming variants were mapped to Vanet before splitting. In phase two, Neysan was kept separate as a held-out subtype. The initially stated extraction count of 184 was corrected by the author to 243 unique images, consistent with the saved notebook outputs; 244 raw Neysan files contained one extra exact copy.

3. Inventory, duplicates, and class distribution

Pillow verified each supported image and then loaded its pixels. Supported formats were JPG, JPEG, PNG, BMP, and WebP. The inventory stored path, dataset, source, original label, normalized label, width, and height. There were no unreadable files in either recorded audit.

SHA-256 file hashes were used for exact duplicate detection. Twenty-four duplicate groups contained 24 extra copies. No duplicate group had conflicting class labels in phase one. In phase two, no exact hash was shared between the Neysan test pool and the other classes, and there were no conflicting expected labels. One copy per hash was retained in the inventory; source files remained intact.

Class label	Meaning used in report	Raw files	Unique files
ambulance	Ambulance	422	408
autobus	Bus	521	520
kamyun	Truck	437	437
kamyunet	Light truck	618	614
minibus	Minibus	438	434
savari	Passenger car	352	352
taxi	Taxi	348	348
vanet	Pickup category, including Neysan in phase one	400	399
Total		3,536	3,512

Table 3. Phase-one recorded class distribution. Transliteration labels are preserved in code and tables to avoid mapping errors.

Raw widths ranged from 117 to 666 pixels, and heights from 156 to 970 pixels. Median width and height were 258 and 278 pixels. The input dimensions therefore varied substantially; a consistent preprocessing rule was necessary.

The largest phase-one unique class was kamyunet (614 images), and the smallest was taxi (348), a ratio of approximately 1.76. Phase two was more imbalanced because only 156 non-Neysan Vanet images remained. Balanced training batches were used to improve class representation.

Scope of the duplicate check

A file hash detects identical bytes. It does not establish that different files depict independent vehicles, independent scenes, or independent capture sessions. Re-encoded copies, near-duplicates, and related frames were not audited in the recorded workflow. Accordingly, this report claims no shared exact file hashes, not complete elimination of every possible form of leakage.

4. Phase-one design and split

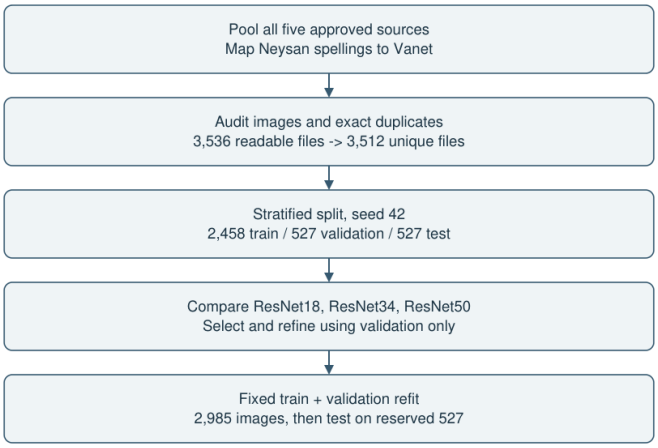


Figure 1. Phase-one workflow. The final test set is reserved during architecture comparison and validation experiments.

Class	Train	Validation	Test	Total
ambulance	285	61	62	408
autobus	364	78	78	520
kamyun	306	66	65	437
kamyunet	430	92	92	614
minibus	304	65	65	434
savari	246	53	53	352
taxi	244	52	52	348
vanet	279	60	60	399
Total	2458	527	527	3512

Table 4. Phase-one stratified 70/15/15 split. All eight classes appear in each subset.

Splitting used a fixed random seed of 42. Paths and hashes were checked for uniqueness and pairwise separation. Manifests, rather than renamed physical source folders, defined the experiment. The test set was not used to choose dropout, loss, sampling, or continuation epochs.

5. Common preprocessing, model, and metrics

5.1. Image preparation

Each image was opened, corrected for EXIF orientation, and converted to RGB. Its aspect ratio was preserved using bilinear resizing inside a 224 x 224 square. The remaining space was filled with RGB (124, 116, 104). Tensors were normalized with mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225). This is a project-specific padding pipeline; it is not claimed to reproduce the default torchvision resize-and-center-crop pipeline.

Training included random horizontal flip with probability 0.5 and color jitter of 0.1 for brightness, contrast, and saturation. Validation and test used deterministic preprocessing. Augmentation remained enabled throughout the selected experiments. No augmentation-off comparison was performed.

5.2. Transfer learning

ResNet uses residual connections to support learning in deeper convolutional networks. The project used ImageNet-pretrained models provided through torchvision, replaced the original classifier with an eight-output layer, and progressively unfroze later feature stages. The chosen ResNet18 head ultimately became Dropout(0.25) followed by Linear(512, 8).

Early feature stages remained frozen during later-stage fine-tuning. BatchNorm running statistics were held fixed by using evaluation mode for BatchNorm modules. Trainable affine parameters in unfrozen stages could still be updated. Dropout was active only during training. The model used its existing average pooling; max-versus-average pooling was not tested.

5.3. Optimization and evaluation

AdamW with weight decay 0.0001 was used. Different learning rates protected pretrained layers while allowing faster adaptation of the classifier. Validation-monitored ReduceLROnPlateau schedules reduced learning rates by a factor of 0.5 with patience 2. Fixed-rate refits were used after validation images became training data.

Measure	Definition / interpretation
Accuracy	Number of correct predictions divided by number of evaluated images.
Precision for class k	TP / (TP + FP): how often a predicted class is correct.
Recall for class k	TP / (TP + FN): how many true examples of that class are recovered.
F1 for class k	2 x precision x recall / (precision + recall).
Macro-F1	Arithmetic mean of the eight class F1 scores in multi-class validation/testing.
Support	Number of true examples belonging to a class in the evaluated subset.

Table 5. Metric definitions. Undefined class precision or recall was set to zero in the classification reports.

6. Phase-one architecture comparison

Each architecture was trained first on its classifier, then on classifier plus layer4, and finally on classifier plus layer3 and layer4. The classifier-only budget was 20 epochs, layer4 fine-tuning used 15 epochs, and layer3+layer4 fine-tuning used 10 epochs. A later stage retained its starting checkpoint if no trained epoch improved the selection criterion.

Architecture	FC only	FC + layer4	FC + layer3 + layer4
ResNet18	92.41%	97.15%	97.53%
ResNet34	94.12%	96.02%	97.34%
ResNet50	96.39%	97.34%	97.34% retained

Table 6. Best retained validation accuracy by stage. ResNet50 layer3+layer4 did not improve on the layer4 checkpoint; the repeated 97.34% is not a new improvement.

Selected architecture checkpoint	Accuracy	Macro-F1	Errors / 527
ResNet18, layer3 + layer4 + FC	97.53%	0.9748	13
ResNet34, layer3 + layer4 + FC	97.34%	0.9744	14
ResNet50, layer4 + FC	97.34%	0.9735	14

Table 7. Evaluation of the selected checkpoint from each architecture on the same validation set.

ResNet18 was chosen because it had the highest recorded accuracy and macro-F1 among these checkpoints and was practical for CPU experiments. Its advantage over the other two models was only one validation image. This is a selection within the performed runs, not proof that ResNet18 is universally better or that deeper models cannot improve.

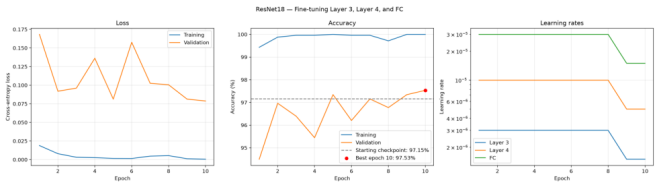


Figure 2. Supplied ResNet18 layer3+layer4 training history. The selected stage epoch was 10.

In ResNet18, the classifier-only model had 4,104 trainable parameters. Unfreezing layer4 increased this to 8,397,832; unfreezing layer3 as well increased it to 10,497,544. Greater adaptation improved this model relative to its frozen-feature baseline.

7. Learning-rate settings and deeper-model histories

Stage	Layer3 initial LR	Layer4 initial LR	FC initial LR
Classifier only	Frozen	Frozen	0.001
Layer4 + FC	Frozen	0.00001	0.0001
Layer3 + layer4 + FC	0.000003	0.00001	0.00003

Table 8. Phase-one initial learning rates. These are initial values, not a claim that every epoch used fixed rates; the notebooks contain validation-loss schedulers.

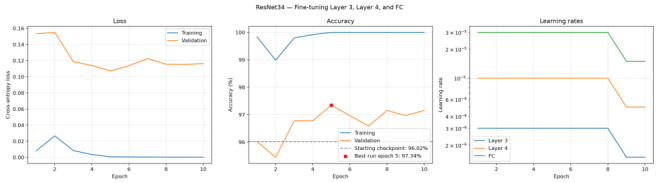


Figure 3. Supplied ResNet34 layer3+layer4 history. The best checkpoint in this stage was epoch 5, with 97.34% validation accuracy.

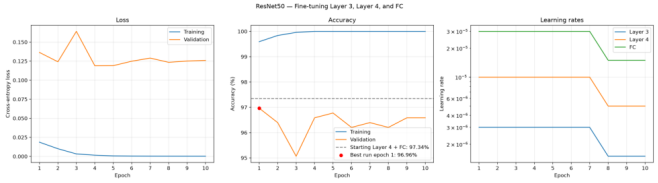


Figure 4. Supplied ResNet50 layer3+layer4 history. The retained 97.34% checkpoint came from the preceding layer4 stage.

Both supplied histories show very high training accuracy without a consistent advantage on validation. Extra depth and additional unfrozen layers increased training cost but did not provide a reliable gain in this comparison. The project therefore concentrated its later experiments on ResNet18 rather than extending the search to ResNet101 or ResNet152.

Training accuracy was measured with augmentation and the current training-mode head. It is not directly equivalent to deterministic validation accuracy. A very high training score alone was not used as a checkpoint-selection rule.

8. ResNet18 refinement: dropout and sampling

Experiment	Best epoch	Validation accuracy	Macro-F1
Dropout 0.00	10	97.72%	0.9768
Dropout 0.25	7	97.91%	0.9784
Dropout 0.50	0 (starting model)		
		97.53%	0.9748
Standard sampling, dropout 0.25	9	98.10%	0.9800
Balanced sampling, dropout 0.25	9	98.29%	0.9819

Table 9. Sequential validation experiments. Each comparison used ten epochs per candidate and the same starting checkpoint within that comparison. Epoch 0 means the starting model was retained.

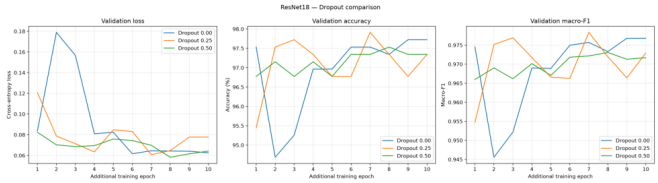


Figure 5. Dropout comparison. Dropout 0.25 was selected; dropout 0.50 did not improve on its initial checkpoint.

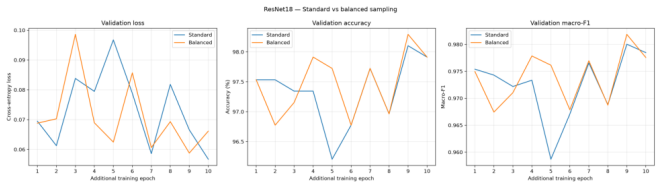


Figure 6. Standard versus balanced sampling. Balanced batches contained two images per class, for 16 images per batch.

Balanced sampling gave equal batch representation to each class. Smaller classes were sampled repeatedly, with augmentation introducing variation. It did not create new independent images. In phase one, 154 balanced batches produced 2,464 image draws per epoch; standard sampling used 2,458 images in 154 batches.

Candidates were selected by macro-F1, then accuracy, then lower loss. Comparisons were single-seed continuation experiments. Additional optimization, random augmentation, and checkpoint selection all contribute to the observed changes. The small differences do not establish statistically confirmed generalization gains from dropout or balancing alone.

9. Loss comparison and additional continuation

Cross-entropy (CE) and binary cross-entropy with logits (BCE) were compared from the same selected balanced-sampling checkpoint. For BCE, an eight-dimensional one-hot target was used. Despite this loss choice, the task remained single-label classification: the output class was always the largest logit.

Run	Selected epoch	Accuracy	Macro-F1
CE continuation	6	98.67%	0.9865
BCE continuation	8	98.67%	0.9868
BCE, five additional epochs	Additional 5	98.86%	0.9890

Table 10. Recorded validation results. CE and BCE each made seven errors at their selected checkpoints; BCE continuation ultimately reduced the count to six.

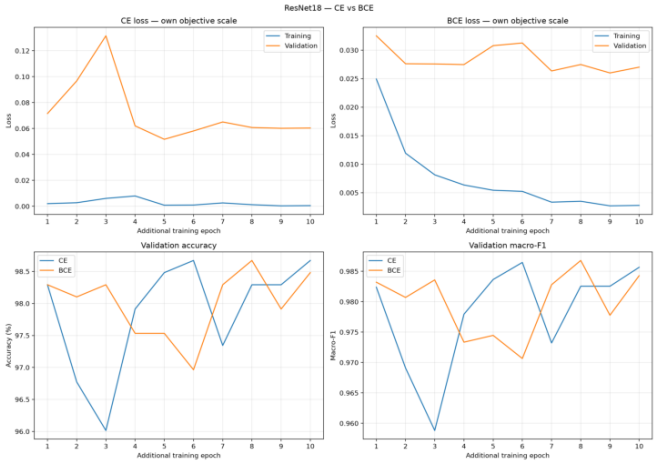


Figure 7. Supplied CE-versus-BCE comparison. The two losses have different scales and should not be ranked by their raw loss values.

The loss comparison scheduler monitored validation macro-F1 rather than raw loss. BCE was selected by a very small macro-F1 difference at equal accuracy. The subsequent five-epoch continuation restored model, optimizer, and scheduler states from the BCE checkpoint. Its restored group rates were 0.0000015, 0.000005, and 0.000015 for layer3, layer4, and FC, respectively.

The final selected validation checkpoint was resnet18_bce_continued_best.pt. Its 98.86% validation accuracy is a model-selection result after multiple experiments on the same validation set. It is not the final held-out test score.

10. Phase-one refit and held-out test

The selected model was refitted on 2,985 training-plus-validation images for a fixed eight additional epochs. The optimizer was fresh AdamW, group rates were 0.0000015 / 0.000005 / 0.000015, and no scheduler was used. Dropout 0.25, BCE, balanced batches, fixed BatchNorm statistics, and augmentation were retained. The saved final-epoch checkpoint was resnet18_trainval_final.pt.

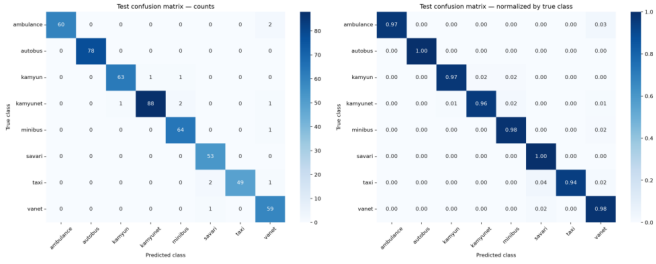


Figure 8. Supplied count and row-normalized confusion matrices on the 527-image phase-one test set.

Class	Precision	Recall	F1	Support
ambulance	1.0000	0.9677	0.9836	62
autobus	1.0000	1.0000	1.0000	78
kamyun	0.9844	0.9692	0.9767	65
kamyunet	0.9888	0.9565	0.9724	92
minibus	0.9552	0.9846	0.9697	65
savari	0.9464	1.0000	0.9725	53
taxi	1.0000	0.9423	0.9703	52
vanet	0.9219	0.9833	0.9516	60

Table 11. Phase-one test classification report. Support totals 527 images.

Final held-out result: 514/527 correct (97.53% accuracy), 0.9746 macro-F1, 0.9746 macro-precision, and 0.9755 macro-recall. Vanet

had the lowest class F1 in this report. The lower test score than the selected validation score is a difference between evaluated subsets and model stages, not evidence by itself of a failed refit.

11. Phase-one error inspection



Figure 9. All 13 supplied phase-one test errors. Images and labels are reproduced from the saved figure; original overlays are part of the supplied image content.

The recorded errors included ambulance-to-Vanet (2), truck/light-truck confusion (2 across both directions), truck-to-minibus (1), light-truck-to-minibus (2), light-truck-to-Vanet (1), minibus-to-Vanet (1), taxi-to-passenger-car (2), taxi-to-Vanet (1), and Vanet-to-passenger-car (1). Similar vehicle shape and limited visible context are plausible explanations, not independently confirmed causes.

Some supplied images include colored rectangles already embedded in their visual content. Those markings were not removed by this project. They could act as dataset-specific cues; performance on differently prepared images may change. This is a reason to seek an independent mentor test rather than claim universal deployment accuracy.

The test errors were inspected after evaluation. No further model settings were selected from these errors in the recorded phase-one workflow. Any future test-driven changes would require a new independent evaluation.

12. All-data model and confidence output

After the held-out phase-one result was recorded, a separate model was continued on all 3,512 unique images for eight fixed epochs. It started from the train-plus-validation checkpoint and kept the same selected dropout, loss, sampler, augmentation, and fixed learning rates. This produced resnet18_all_data_final.pt.

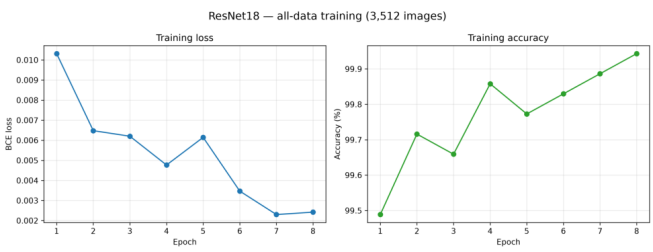


Figure 10. Supplied phase-one all-data training history. Final sampled training accuracy was 99.94%; this is not an independent test result.

The all-data checkpoint has no remaining local holdout. The 97.53% result belongs to resnet18_trainval_final.pt, not to this later all-data model. An independent mentor dataset is needed to evaluate the all-data checkpoint.

12.1. Confidence definition

For the selected BCE model, prediction uses the class with the largest logit z. The confidence score is sigmoid(z) for that predicted class, multiplied by 100 for display. Sigmoid scores across classes do not have to sum to one. A score of 99% should not be interpreted as a calibrated 99% probability that the prediction is correct.

The earlier architecture-comparison notebook used softmax confidence for CE checkpoints. Later BCE prediction and final-test code used sigmoid scores. Those two confidence definitions should not be treated as interchangeable calibration measurements.

12.2. Portable evaluation

The mentor evaluator reconstructs the architecture with weights=None, loads saved model weights and preprocessing settings, and evaluates CPU batches. It needs the Python script, checkpoint, suitable dependencies, and labeled images. It does not require the original notebooks or a download of ImageNet weights. Exact training copies are excluded and recorded; supplied and evaluated image counts are reported separately.

The provided archives contain notebook code and output figures, but not model-weight binaries. The report reviews saved results and implementations; it does not claim an independent rerun of inference or training.

13. Phase two: Neysan held out from training

Phase two tests subtype generalization. Neysan examples were separated from Vanet and assigned the role neysan_test. Their expected prediction remained Vanet; no ninth output class was introduced. The raw audit found 244 Neysan files, of which 243 were unique. The other 3,269 unique images formed the eight-class pool.

Class	Train (80%)	Validation (20%)	Pool
ambulance	326	82	408
autobus	416	104	520
kamyun	350	87	437
kamyunet	491	123	614
minibus	347	87	434
savari	282	70	352
taxi	278	70	348
vanet	125	31	156
Total	2615	654	3269

Table 12. Phase-two stratified split, seed 42. The 243 Neysan images were outside both columns and outside the later refit pool.

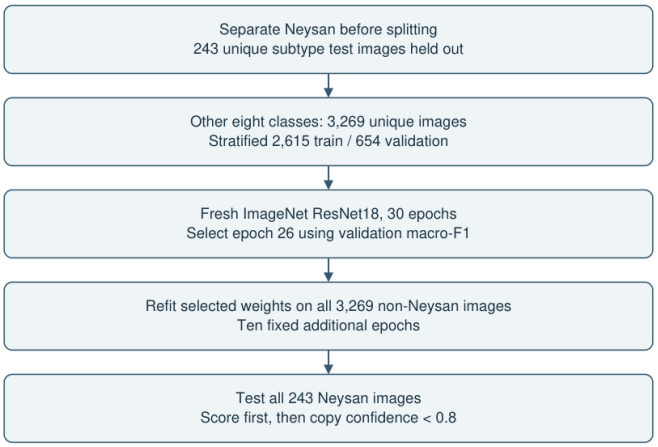


Figure 11. Neysan-holdout workflow. Phase-one vehicle-trained checkpoints were not reused because they had already seen Neysan.

The remaining Vanet pool was small. Two images per class in each balanced batch ensured representation of Vanet during optimization. The 125 training examples were revisited with augmentation. This reduces unequal batch frequency but cannot replace missing visual diversity.

14. Phase-two training and validation selection

A fresh ImageNet-pretrained ResNet18 was initialized with Dropout(0.25) and a new eight-class head. Only layer3, layer4, and FC were trainable. The budget was 30 epochs with BCE one-hot targets, AdamW, weight decay 0.0001, batch size 16, and fixed BatchNorm statistics. Initial group rates were 0.000003, 0.00001, and 0.001. The new classifier used a higher rate because its weights had not yet been trained on these classes.

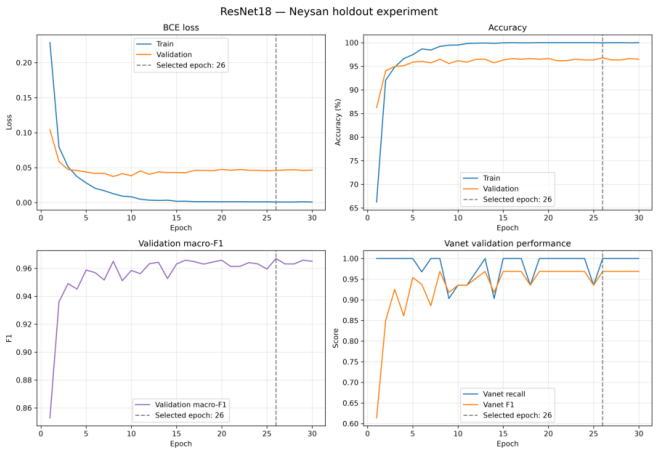


Figure 12. Supplied phase-two training curves, including Vanet validation recall and F1. The selected checkpoint was epoch 26.

Class	Precision	Recall	F1	Support
ambulance	1.0000	0.9878	0.9939	82
autobus	0.9904	0.9904	0.9904	104
kamyun	0.9651	0.9540	0.9595	87
kamyunet	0.9672	0.9593	0.9633	123
minibus	0.9651	0.9540	0.9595	87
savari	0.9189	0.9714	0.9444	70
taxi	0.9706	0.9429	0.9565	70
vanet	0.9394	1.0000	0.9688	31

Table 13. Epoch-26 validation report. Overall accuracy was 96.79% and macro-F1 was 0.9670.

Selection ranked macro-F1 first, accuracy second, and lower loss third. The scheduler monitored validation macro-F1. All 31 non-Neysan validation Vanet images were recovered, but that result alone did not predict performance on the held-out Neysan subtype. Recorded training time was 208.6 minutes.

15. Phase-two refit and Neysan test result

The epoch-26 model was continued on training plus validation: 3,269 non-Neysan images, including all 156 remaining Vanet images. The final budget was fixed at ten additional epochs before Neysan evaluation. A fresh AdamW optimizer used the saved reduced rates: 0.00000009375 for layer3, 0.0000003125 for layer4, and 0.00003125 for FC. No validation scheduler or best-epoch selection was used during this refit.

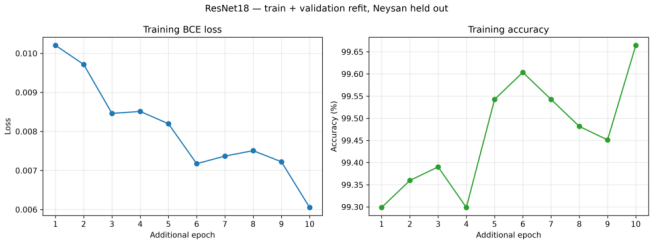


Figure 13. Supplied phase-two fixed-budget refit history. Training loss fell from 0.010207 to 0.006054. Final sampled training accuracy was 99.66%.

Balanced refit epochs contained 205 batches and 3,280 image draws. Refit training lasted 75.3 minutes. The model saved after epoch 10 was resnet18_neysan_trainval_final.pt. All 243 unique Neysan test hashes were disjoint from its training hashes.

Prediction on Neysan	Images	Share of 243
vanet (correct)	228	93.83%
kamyun	6	2.47%
kamyunet	5	2.06%
minibus	4	1.65%
Total	243	100% (rounding applies)

Table 14. Neysan prediction distribution. The correct target for every image was Vanet.

Subtype recognition accuracy was 228/243 = 93.83%. On this single-true-class test, Vanet recall is the same 93.83%. Vanet precision is 1.0 because there are no non-Vanet true examples that could produce Vanet false positives. The resulting Vanet F1 is 0.9682, which must not be confused with eight-class macro-F1.

The phase-one 97.53% score and phase-two 93.83% score are not a direct architecture comparison. They use different training pools, splits, model histories, and test populations. Phase two specifically asks whether an excluded subtype is assigned to its intended parent class.

16. Confidence threshold and human-review routing

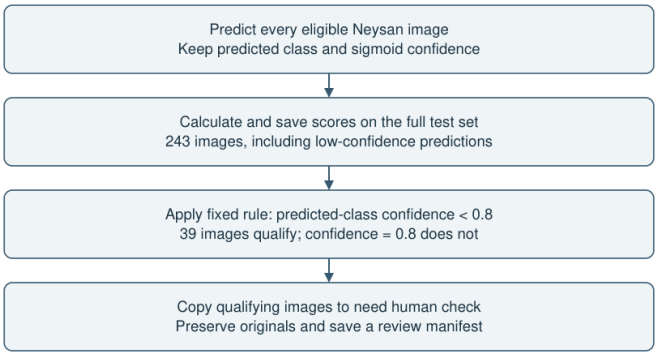


Figure 14. Review routing occurs after scoring. The confidence threshold does not remove difficult images from the reported test denominator.

Predicted-class confidence	Correct	Incorrect	Total
Below 80%	29	10	39
At least 80%	199	5	204
Total	228	15	243

Table 15. Confidence routing outcomes. Correctness is assessed against expected Vanet labels, independently of the threshold.

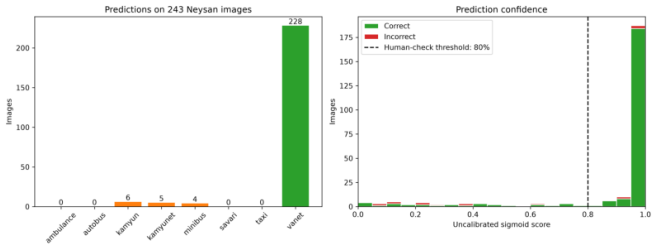


Figure 15. Supplied prediction-count and confidence-distribution summary for the Neysan test.

The review workload was 39/243 = 16.05% of the test set. The threshold captured 10/15 = 66.67% of observed errors. Ten of the 39 flagged images were errors (25.64%), while 29 were correct. Among the 204 images above or at the threshold, five were wrong; their observed accuracy was 199/204 = 97.55%. These are descriptive outcomes of this test, not estimates of guaranteed future reliability.

Files were copied, not moved, into a folder named need human check. Hash-based names prevent collisions. A CSV manifest connects each copy with its original path, prediction, score, and correctness. Human corrections or post-review accuracy were not recorded, so no assisted-performance claim is made.

17. Neysan errors and visual interpretation



Figure 16. All 15 supplied Neysan errors, sorted by predicted-class confidence. Ten are below the review threshold and five remain at or above it.

The error destinations were trucks, light trucks, and minibuses. Visual similarity, occlusion, camera angle, background context, or annotation ambiguity are possible explanations. The supplied examples support inspection, but do not isolate a causal mechanism. The five confident errors show why a confidence threshold is a review aid rather than a guarantee.

Neysan was held out to evaluate generalization, so these images should not be fed back into model selection while preserving the same test claim. Any subsequent training based on their errors would require a new independent test set.

18. Mentor delivery and reproducibility

Two standalone scripts were supplied: mentor_evaluation.py for labeled eight-class evaluation, and neysan_mentor_evaluation.py for the subtype experiment. The latter requires a folder containing only Neysan images, including any nested subfolders; every image is assigned expected_class = vanet. It must not be pointed at a mixed-class dataset.

Deliverable	Role
neysan_mentor_evaluation.py	Loads the phase-two model, excludes exact training overlaps, predicts and scores images, then copies low-confidence cases.
resnet18_neysan_trainval_final.pt	Required checkpoint with model weights, class mapping, configuration, training hashes, and stage metadata.

Deliverable	Role
requirements.txt / environment record	Dependencies: PyTorch, torchvision, Pillow, pandas, and scikit-learn for the evaluator.
Mentor Neysan image folder	Only Neysan examples; no original project data are required.
Empty output folder	Outside the image input directory to prevent review copies being re-read as test inputs.

Table 16. Phase-two mentor handoff requirements. Model checkpoints were not included in the submitted codes.zip and images.zip report inputs.

19. Test on mentors dataset

On the day I presented my project, the mentors gave us additional image datasets for testing. I evaluated the best model from phase one and the best model from phase two on folders 0, 2, 3, and 4, named test0, test2, test3, and test4.

These folders use eight class labels, but differ in the presence of Neysan images and in how some vehicles are labeled. The dataset explanation supplied with the test data describes them as follows:

test0 (folder 0): Eight classes, with no Neysan images.

test2 (folder 2): Eight classes. Neysan images are included in the Vanet folder.

test3 (folder 3): Eight classes. Neysan images are distributed among other class folders.

test4 (folder 4): Eight classes. Neysan and other pickup images are grouped under Vanet; Truck (kamyun) and Light truck (kamyunet) images are incorrectly categorized.

Excluded datasets: test1 and test1_2 were not used because they contain nine classes, while both models predict eight classes. In test1, Neysan is a separate class. The supplied explanation also identifies incorrect Truck and Light truck categorization in test1_2.

19.1. Test results

Test folder	Phase 1 (%)	Phase 2 (%)
0 (test0)	95.01	95.01
2 (test2)	95.26	94.95
3 (test3)	87.85	87.75
4 (test4)	84.95	84.05

Table 17. Author-reported accuracy of the best phase-one and phase-two models on the mentor test datasets.

Both models achieved 95.01% on test0. Phase one scored slightly higher on test2, test3, and test4. The lower scores on test3 and test4 should be read alongside their labeling differences. These results alone do not establish how much of the decrease comes from labeling or image difficulty.

20. Conclusion

This project developed and evaluated a CPU-based, eight-class vehicle classifier through source pooling, manual correction, exact deduplication, stratified splitting, and transfer learning. ResNet18 was selected from the recorded comparisons and refined with dropout 0.25, balanced batches, and BCE loss.

The selected phase-one validation model reached 98.86% accuracy and 0.9890 macro-F1. The fixed training-plus-validation refit achieved 97.53% accuracy and 0.9746 macro-F1 on 527 held-out images. A separate phase-two model trained without Neysan examples correctly assigned 228 of 243 held-out Neysan images to Vanet (93.83%). The confidence threshold flagged 39 images and captured 10 of the 15 errors.

The best models from both phases were also tested on the additional datasets provided by the mentors on presentation day. Across folders 0, 2, 3, and 4, phase-one accuracy ranged from 84.95% to 95.26%, and phase-two accuracy ranged from 84.05% to 95.01%. Both models reached 95.01% on the dataset without Neysan images. These external results extend the evaluation beyond the original project splits and document performance under different Neysan labeling arrangements.

The original Neysan-only result measures recognition of a held-out subtype, while the mentor tests evaluate mixed eight-class datasets. Their scores therefore describe different evaluation settings. More diverse images and consistent class labeling would help strengthen future evaluation.